

the derivative operators to the left of the θ functions in Eq. (6.2.8)

$$\Delta_{\ell m}(x, y) = P_{\ell m}^{(L)} \left(-i \frac{\partial}{\partial x} \right) \Delta_F(x - y) + \delta(x^0 - y^0) P_{\ell m}^{(1)} (-i \nabla) [\Delta_+(x - y) - \Delta_+(y - x)], \quad (6.2.12)$$

where Δ_F is the 'Feynman propagator'

$$-i\Delta_F(x) \equiv \theta(x)\Delta_+(x) + \theta(-x)\Delta_+(-x). \quad (6.2.13)$$

However, for $x^0 = 0$ the function $\Delta_+(x)$ is even in $\mathbf{x}$, since a change $\mathbf{x} \rightarrow -\mathbf{x}$ in Eq. (6.2.9) can be compensated by a change $\mathbf{p} \rightarrow -\mathbf{p}$ in the integration variable. We can therefore drop the second term in Eq. (6.2.12), and write simply

$$\Delta_{\ell m}(x, y) = P_{\ell m}^{(L)} \left(-i \frac{\partial}{\partial x} \right) \Delta_F(x - y). \quad (6.2.14)$$

It will be most useful to use the expression of the Feynman propagator as a Fourier integral. The step functions in Eq. (6.2.13) have the Fourier representation*

$$\theta(t) = \frac{-1}{2\pi i} \int_{-\infty}^{\infty} \frac{\exp(-ist)}{s + i\epsilon} ds. \quad (6.2.15)$$

This can be combined with the Fourier integral (6.2.9) for $\Delta_+(x)$. We introduce new integration variables, $\mathbf{q} \equiv \mathbf{p}$, $q^0 = p^0 + s$ in the first term of Eq. (6.2.13), yielding

$$\begin{aligned} -i\Delta_F(x) &= -\frac{1}{2\pi i} \int d^3q \int_{-\infty}^{\infty} dq^0 \frac{\exp(i\mathbf{q} \cdot \mathbf{x} - iq^0 x^0)}{(2\pi)^3 \sqrt{2\sqrt{\mathbf{q}^2 + m^2}}} \\ &\quad \times \left[\left(q^0 - \sqrt{\mathbf{q}^2 + m^2} - i\epsilon \right)^{-1} + \left(-q^0 - \sqrt{\mathbf{q}^2 + m^2} - i\epsilon \right)^{-1} \right]. \end{aligned}$$

Combining denominators and adopting a four-dimensional notation, we have simply

$$\Delta_F(x) = (2\pi)^{-4} \int d^4q \frac{\exp(iq \cdot x)}{q^2 + m^2 - i\epsilon}, \quad (6.2.16)$$

where $q^2 \equiv \mathbf{q}^2 - (q^0)^2$. (In the denominator we have replaced $2\epsilon\sqrt{\mathbf{q}^2 + m^2}$ with ϵ , because the only important thing about this quantity is that it

* To prove this, note that if $t > 0$ then the contour of integration can be closed with a large clockwise semi-circle in the lower half-plane, so the integral picks up a contribution of $-2\pi i$ from the pole at $s = -i\epsilon$. If $t < 0$ then the contour can be closed with a large counter-clockwise semi-circle in the upper half-plane, where the integrand is analytic, giving an integral equal to zero.